Training Details:

Course Name	Copilot in Microsoft Fabric: Data Engineering
Level	Intermediate
Duration	8 Hrs
Participant background (Prerequisites)	Data Engineering Concept in Microsoft Fabric

Topics	**Number of Days**
Introduction to Copilot in Microsoft Fabric	1
Copilot in Data Factory	
Copilot in Data Warehouse	
Copilot for Real-Time Intelligence	
Copilot in Data Science and Engineering Notebooks	
Copilot Consumption and Data Agent in Fabric	
Total	**1**

Copilot in Microsoft Fabric: Data Engineering

Prerequisite:

- Data Engineering Concept in Microsoft Fabric

Introduction to Copilot in Microsoft Fabric

- Overview of Copilot in Microsoft Fabric
  - Role of Copilot in enhancing productivity and insights.
  - Integration across Data Factory, Data Warehouse, Real-Time Intelligence, and Notebooks.
- Enabling Copilot Features
  - Tenant settings and region availability.
  - Licensing requirements and supported SKUs.

Copilot in Data Factory

- Dataflow Gen2 Enhancements
  - Natural language to transformation steps.
  - Query summarization and generation.
- Data Pipeline Automation
  - Pipeline generation from natural language descriptions.
  - Error message assistance and pipeline summarization.

Hands-On Lab: Create a dataflow and pipeline using Copilot.

Copilot in Data Warehouse

- Natural Language to SQL
  - Generating T-SQL queries from plain English prompts.
  - Understanding schema-aware query generation.
- Code Assistance Features
  - Code completion and quick actions (Fix and Explain).
  - Best practices for prompt crafting.

Hands-On Lab: Interact with Copilot in the Data Warehouse editor.

Copilot for Real-Time Intelligence

- KQL Query Generation

    - Translating business questions into Kusto Query Language (KQL).
    - Dynamic query refinement and follow-up interactions.
- Use Cases
    - Real-time analytics scenarios and applications.

Hands-On Lab: Analyze real-time data using Copilot-generated KQL queries.

Copilot in Data Science and Engineering Notebooks

- Notebook Integration
    - Working with Lakehouse tables, Power BI Datasets, and DataFrames.
    - Generating code snippets for data analysis and visualization.
- Chat-magics in Fabric Notebook
- AI-Assisted Development
    - Model type suggestions and error resolution.
    - Automatic documentation and code summarization.

Hands-On Lab: Develop a data analysis notebook utilizing Copilot features.

Copilot Consumption and Data Agent in Fabric

- Copilot Consumption Metrics
    - Token-based billing model and capacity unit (CU) consumption rates.
    - Monitoring usage through the Fabric Capacity Metrics app.
- Data Agent Overview
    - Role of data agents in facilitating Copilot interactions.
    - Configuration and management within Fabric.
- Best Practices
    - Optimizing Copilot usage for cost and performance.

Hands-On Lab: Create a data agent in the Data Warehouse.